War or Peace? Forecasting Military Escalation through Propaganda Posters

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[Git Repository]

1 Introduction

Can we anticipate military escalation from major power? As one of the central questions in political science, the answer concerns not only a broad scholarship in international relations and political violence, but also and diplomatic policymaking.

As with other forecasting tasks, machine learning models have been widely deployed on this question. Starting from the ground-breaking work of Beck, King and Zeng (2000), which used neural networks to predict militarized interstate disputes, time-series models (Brandt, Freeman and Schrodt, 2011), random forests (Muchlinski et al., 2016), and gradient boosting (Hegre et al., 2021) were successively introduced.

A second line of work focuses on the training data itself, moving from structural variables such as conflict history, population, development, and geography (Hegre, Nygård and Landsverk, 2021) to unstructured data. A representative direction is the rise of text-as-data. Chadefaux (2014) first introduced news text into war early-warning, while Mueller and Rauh (2018) further employed topic models to extract information from newspaper articles, systematically incorporating media information into conflict forecasting. More recently, Wong (2026) extended this line of research by exploring the predictive value of official rhetoric.

[1]Team ValueError, supervised by TA Yuhan Luo. All authors contributed equally and are thus listed alphabetically.

[2]Overleaf Word Count: 1463

[3]ChatGPT-6 was used to check grammar and proofread after the initial draft was completed. The conversation history is available [here].

Following this line of research, this project seeks to advance military escalation forecasting by extending the analysis of unstructured data beyond text. As a form of tacit knowledge (Polanyi, 1966), images are often more effective in shaping emotions (Brader, 2005; Slovic et al., 2017). Powell et al. (2015) shows that when text and images appear together, textual frames shape opinions, while visual frames drive behavioral intentions. Images may therefore play a distinctive role in wartime mass mobilization and legitimacy-building (Mattingly and Yao, 2022), offering a potential avenue for improving military escalation forecasts.

Building on these, we compile an original dataset of 5,619 Chinese propaganda posters spanning the 1950s to the 2010s. By extracting visual and textual features, we explore their potential to improve forecasts of military escalation.

2 Data

Two main datasets were applied: Posters collected from ChinesePosters.net through a WebCrawler, and the Militarized Interstate Disputes (MIDs) dataset, from the Correlates of War Project (Palmer et al., 2022).

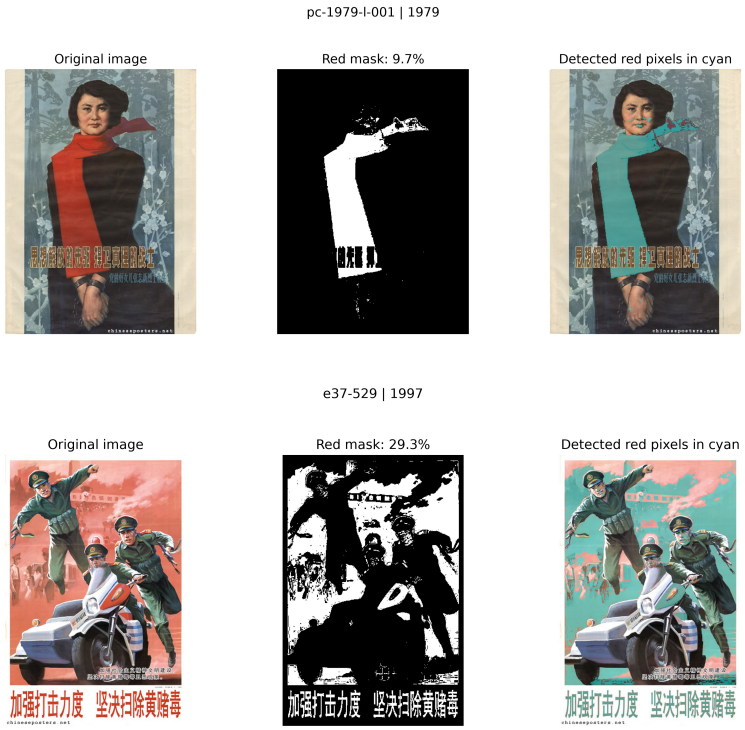
2.1 Propaganda Posters

We crawled 5,619 unique poster records from the Chinese Posters Foundation. The collected records contain poster images with Chinese and English titles, publication dates, publishers, designers, physical dimensions, website descriptions, themes, and tags. After parsing publication dates, we retain 4,752 posters with unambiguous publication years between 1949 and 2013.

For data preprocessing, we first deal with color features by applying EXIF orientation correction and converted images with embedded color profiles to sRGB. We calculate mean RGB values, mean HSV saturation, and CIELAB lightness statistics for each poster using scikit-image (van der Walt et al., 2014). Meanwhile, we identify red pixels through threshold-based segmentation in HSV color space (OpenCV, N.d.). Specifically, we define red pixels as having a hue within 15° of

red, saturation ≥ 0.4, and value ≥ 0.15. This procedure is illustrated in following example:

Figure 1: Examples of red-pixel detection



After extracting the color features, we standardize the recorded physical dimensions by correcting clear formatting inconsistencies, and after that We derive the shorter side, longer side, printed area, and aspect ratio. This process retains some unsupported or ambiguous formats which we save as missing values. We also extract Chinese-title availability and Chinese character counts. For designers, we count identified named entities and create an indicator for multiple entities.

Website description lengths, theme and tag counts, and selected subject indicators form a separate feature group. While these retrospective annotations provide additional information, they may also introduce data leakage in temporal prediction. We therefore conservatively retain only two military-related tags.

We retain observations with missing feature values rather than dropping them. To better align with the target variable, we aggregate features into annual means.[4]

[4]For binary indicators, annual means represent proportions among non-missing records.

2.2 Military Escalation

Following the established literature on military escalation prediction, we use the Dyadic Militarized Interstate Disputes dataset, version 4.03. (Beck, King and Zeng, 2000; Palmer et al., 2022). We focus on records involving China (country code 710) from 1950 to 2014.

Our target variable is the annual number of distinct disputes in which China is involved. As a single dispute can involve multiple countries, we count each dispute only once per year, and we count ongoing disputes in every year they remain active. The outcome thus captures China's annual participation in militarized disputes. Matching poster features from year t to dispute counts in year $t + 1$ yields 65 annual observations.

2.3 Dataset Overview

The final dataset contains 65 rows and 125 columns, including identifiers, aggregated features, data-availability indicators, the target variable, and the split label. The unit of observation is a calendar year. Poster features from year t are matched to a single count target: the number of distinct militarized interstate disputes involving China that are active in year $t+1$. Detailed feature descriptions are provided in the appendix and in the data documentation in our Git Repository.

Table 1: Dataset overview

	Years covered	Observations
Collected Poster Records	Before Year Filtering	5,619
Eligible Propaganda posters	1949–2013	4,752
Military escalation (MID)	1950–2014	65
Matched Annual X/Y Dataset	X:1949-2013; Y:1950-2014	65

3 Exploratory Data Analysis

Given the temporal structure of the data, we chronologically divide the final 65 observations into 52 training and validation observations (poster years 1949–2000) and 13 held-out test observations (2001–2013). All EDA in this section uses only the training and validation data.

3.1 From 25 Candidates to 12 Features

To ensure broad representation across different types of information, we begin by identifying 25 theoretically informed candidate features from the original dataset, covering three main categories:

- **Visual:** Features directly extracted from poster images, including red, green, and blue color measures, lightness, saturation, and other visual characteristics.
- **Metadata:** Information associated with the posters but not directly embedded in the images, including titles, designer characteristics, and publisher attributes.
- **Archive:** Subjective annotations added by the collectors or archivists. To minimize potential biases introduced by these retrospective classifications, we conservatively selected only two military-related indicators.

Given that 25 features may lead to overfitting with limited training observations, we examined feature variability, pairwise Pearson correlations, conceptual overlap, and relevance to the research question to reduce the feature set. We retained one representative from several highly overlapping groups. The full correlation matrix and the 13 exclusion decisions appear in Appendix A1.

We also excluded nearly constant or particularly sparse name-based indicators. These exclusions were exploratory choices, and we will carefully consider how to incorporate these features and better handle omitted variables in the subsequent model training stage.

Table 2 lists the final provisional selection.

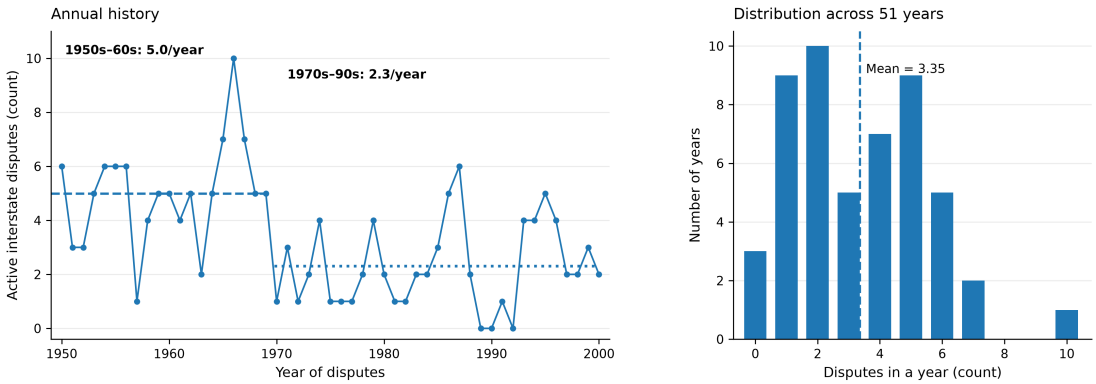
Table 2: Selected features for the proposed regression model.

Category	Feature	Description	Selection Rationale
Visual	red_ratio_mean	Mean proportion of red pixels	Captures red-color prevalence
Visual	saturation_mean_mean	Mean color saturation	Measures color vividness
Visual	lightness_mean_mean	Mean image lightness	Represents overall brightness
Visual	lightness_std_mean	Mean within-image lightness standard deviation	Captures brightness variation
Visual	size_area_cm2_mean	Mean printed area (cm^2)	Represents physical poster size
Visual	size_aspect_ratio_mean	Mean poster aspect ratio	Captures poster shape
Metadata	title_han_count_mean	Mean Chinese character count in titles	Measures title length
Metadata	designer_has_identified_mean	Proportion with identified designers	Captures designer attribution
Metadata	designer_org_name_marker_mean	Proportion with organizational designer-name markers	Captures organizational attribution
Metadata	publisher_military_name_marker_mean	Proportion with military publisher-name markers	Captures military-related publishing
Archive	archive_theme_military_mean	Proportion with military-theme labels	Captures military-themed content
Archive	archive_tag_military_personnel_mean	Proportion with military-personnel tags	Captures military personnel references

3.2 Visualizations and Explanatory Findings

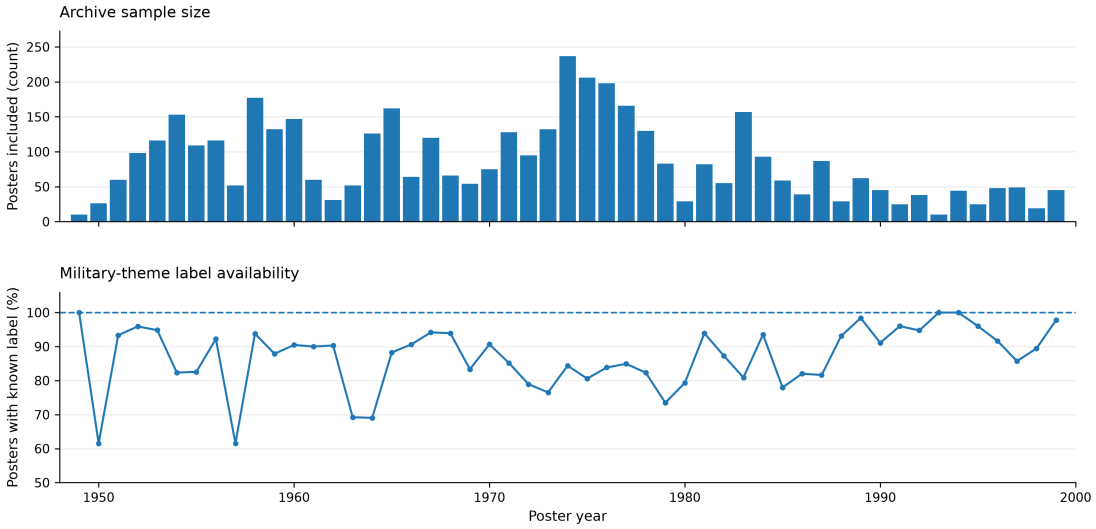
As shown in Figure 2, the annual number of active MIDs varies considerably across the study period. Notably, the average annual dispute count declined from 5.0 during 1950–1969 to 2.3 during 1970–1999. This historical variation suggests that the underlying relationships may change over time, motivating chronological rather than random splits.

Figure 2: Decline in Military Disputes Involving China



Note: The left panel displays annual dispute counts, while the right panel shows their distribution across the training set. Average counts were higher during 1950–1969 than during 1970–1999.

Figure 3: Declining Poster Production



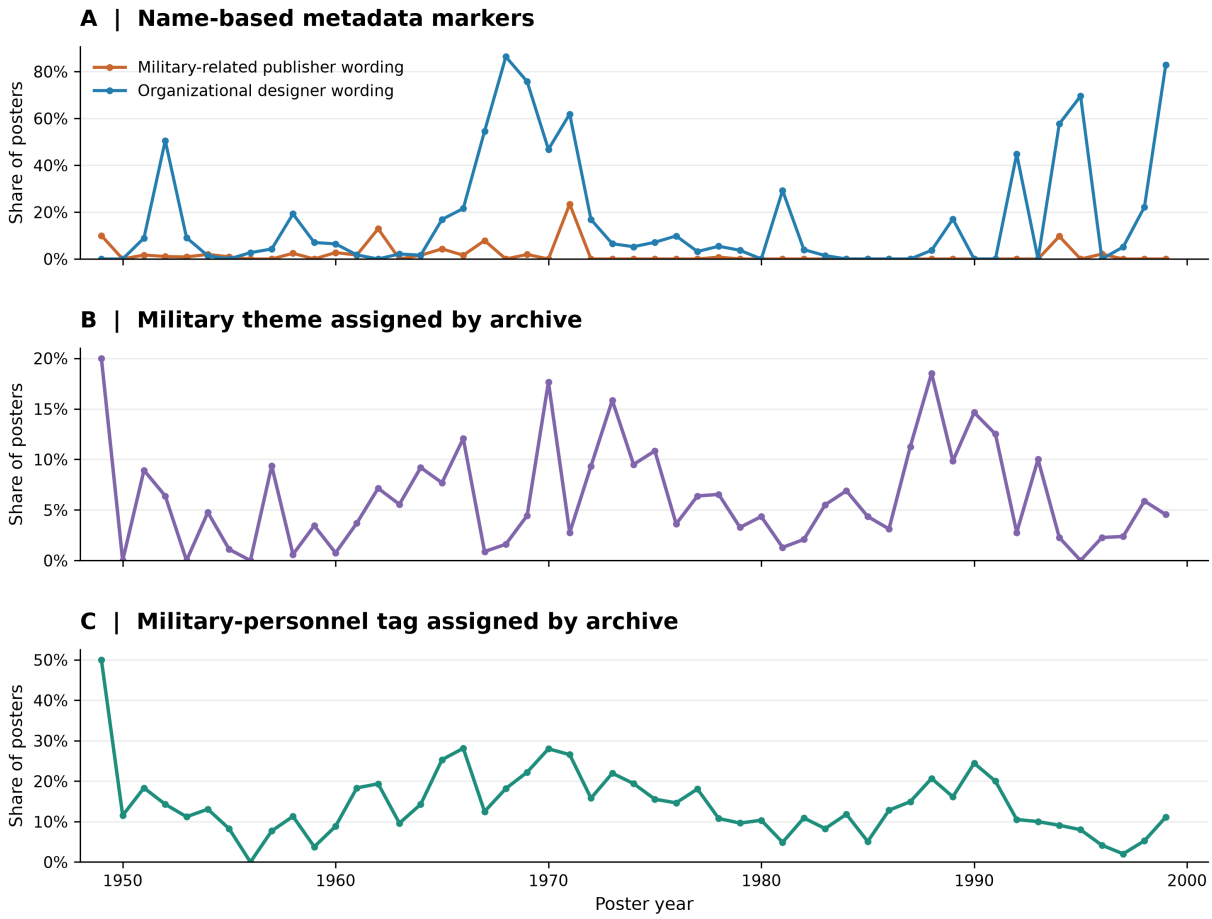
Note: Yearly number of eligible posters and percentage with a valid military-theme annotation.

Figure 3 suggests that archive availability is uneven. Overall, however, a potential positive correlation can be observed: as China's involvement in militarized disputes declined, the production

of military-related posters also appears to have decreased.

Military markers show broadly similar patterns. Figure 4 displays two indicators: military publishers and organizational designers tag, and two archival annotations: military themes and personnel tag. Their peaks coincide with or precede heightened disputes, indicating propaganda may signal military escalation via mass mobilizing.

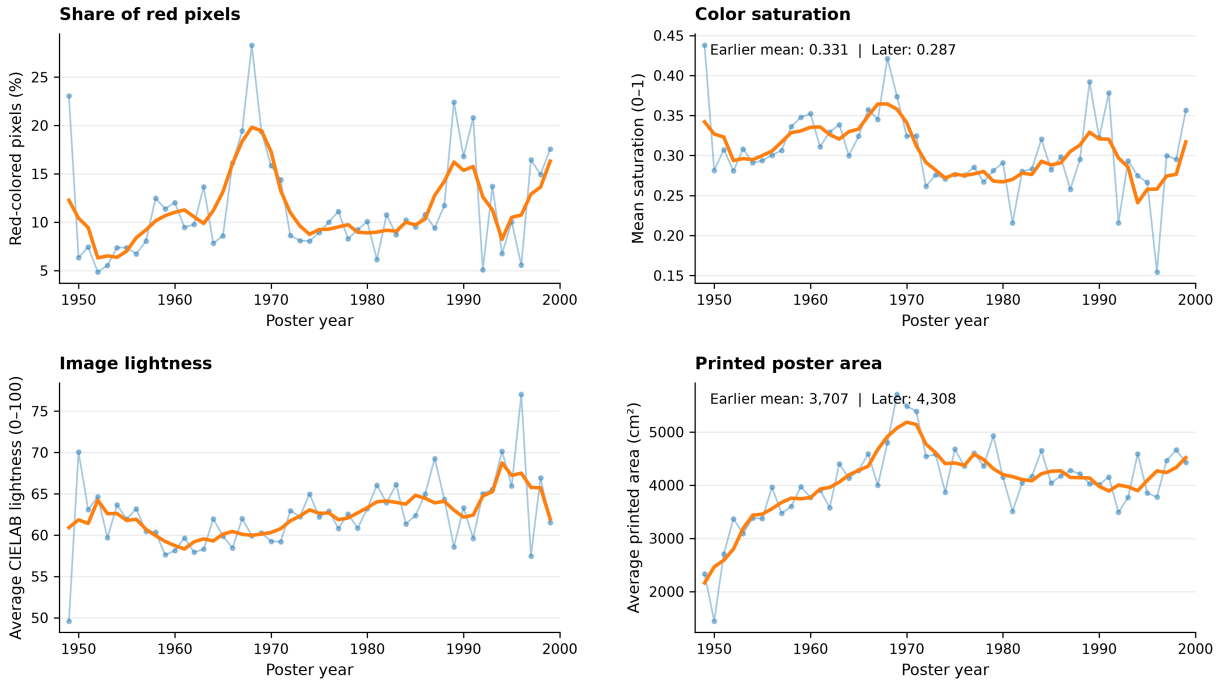
Figure 4: Coinciding Peaks in Military-Related Poster Indicators and Military Disputes



Note: Three aligned time-series panels. Vertical scales reflect shares among records with known values and may differ between panels.

Visual properties, however, present more mixed evidence. Figure 5 compares red-pixel share, saturation, lightness, and printed area. Earlier posters generally exhibit higher saturation, while later posters tend to have larger printed areas. Their relationship with military escalation, however, leaves room for more nuanced exploration.

Figure 5: Distinct Visual Styles Across Historical Periods but Mixed Evidence in Disputes



*Note:*Four annual visual summaries. Moving averages are descriptive. Faint curves indicate observed yearly means; thick curves show centered five-year descriptive averages. Units are shown on each axis.

4 Problem Description and Potential Impact

As discussed above, this project is essentially a regression task with a continuous outcome, which can be expressed as:

$$\text{Military Escalation}_{t+1} = f(\text{Visual Features}_t, \text{Poster Metadata}_t, \text{Archive Labels}_t) \quad (1)$$

Our poster dataset provides the empirical foundation for this project. Previous studies of this collection have primarily focused on art history (Landsberger, 1994, 2013), with limited applications in social science (Chen et al., 2023). To our knowledge, no study has systematically applied computational methods to this collection, particularly for forecasting.

For the outcome variable, we use the established MIDs dataset (Ward, Siverson and Cao, 2007; Carroll and Kenkel, 2019) rather than constructing an arbitrary alternative to better engage the

scholarship.[5] Existing studies have demonstrated the predictive potential of neural networks (Beck, King and Zeng, 2000) and unstructured diplomatic correspondence (Boussalis et al., 2022). Building on this scholarship, we seeks to explore the broader potential of multimodal data, particularly visual information, for improving military escalation forecasts.

Following the methods covered in the course, we will first estimate a linear regression model as an interpretable baseline. Drawing on the existing literature (Muchlinski et al., 2016; Hegre et al., 2021), we will further consider widely used machine learning models, including random forests and gradient boosting, to assess whether more flexible approaches improve overall performance.

Three additional technical considerations are particularly important:

- **Time-series structure:** Observations may violate the i.i.d. assumption. We therefore use chronological splits to prevent future information leakage.
- **Choice of time window:** We initially predict one year ahead but acknowledge the arbitrariness. Alternative windows will be considered for robustness during model training.
- **Missing values:** We will explore appropriate imputation strategies based on methods introduced later in the course.

This project is an exploratory contribution to the long-standing challenge of war forecasting. Its primary audience is academic researchers in international relations and political communication. By examining whether propaganda posters improve predictive performance, we seek to demonstrate the potential of visual information as a non-traditional data source for conflict forecasting.

The project may also inform foreign policymaking. Although preliminary, the model could potentially help policymakers assess China's foreign policy through newly released posters. In particular, identifying which visual and textual features precede military escalation may provide additional insights to support broader policy assessments.

[5] This approach has been discussed with the TA.

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Appendix

A Feature Selection and Supporting Diagnostics

A.1 Excluded Variables and Selection Rationales

Table 3: Excluded candidate features and reasons for removal.

Excluded Feature	**Reason for Exclusion**
`r_mean_mean`	Overlaps with other retained color and brightness measurements.
`g_mean_mean`	Highly correlated with mean lightness ($r \approx 0.981$).
`b_mean_mean`	Strongly correlated with green intensity and mean lightness.
`lightness_p90_p10_mean`	Highly correlated with lightness standard deviation ($r \approx 0.958$).
`size_short_cm_mean`	Highly correlated with poster area ($r \approx 0.962$).
`size_long_cm_mean`	Highly correlated with poster area ($r \approx 0.910$).
`title_has_chinese_mean`	Nearly constant across development observations.
`multi_designer_mean`	Lower priority for the compact, interpretable feature set.
`publisher_has_name_mean`	Limited variation across development observations.
`designer_collective_name_marker_mean`	Sparse indicator; zero in 40 of 51 development years.
`publisher_press_name_marker_mean`	Lower priority for the research-focused feature set.
`publisher_party_name_marker_mean`	Nearly constant; zero in 50 of 51 development years.
`publisher_education_name_marker_mean`	Sparse and sensitive to an unusual observation; zero in 45 of 51 years.

A.2 Pairwise Feature Correlation Matrix

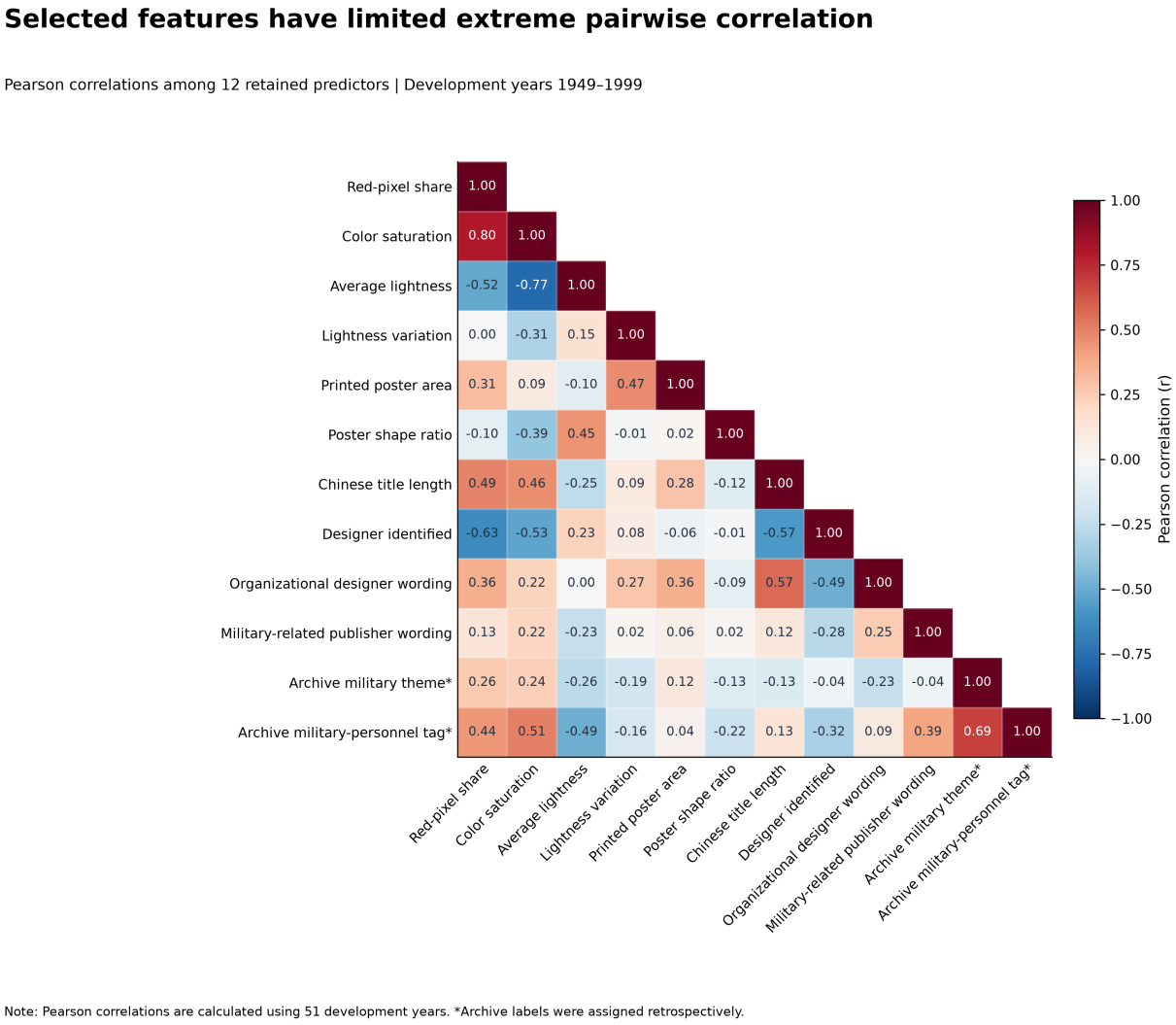


Figure A1: Pearson correlations among the retained predictors, using only development observations. The upper triangle is omitted because the matrix is symmetric. The largest off-diagonal absolute correlation is 0.801.